OCHMO Office of Research Assurance

Human Subject Research Protections Standard Operating Procedures (SOP)

SOP- 800: ORA Quality Improvement and Outreach Program (QIOP)

Number	Effective Date	Version Number	Previous Versions
800	[Effective Date]	v.1.1	v.1

Purpose

The purpose of the Office of Research Assurance (ORA)'s Quality Improvement and Outreach Program (QIOP) is to facilitate research and support compliance with relevant federal regulations, NASA policies, and Institutional Review Board (IRB) approved procedures. Working collaboratively with Principal Investigators (PI) the program aims to proactively identify and address potential compliance concerns through follow-up and targeted education. SOP-800 outlines the program objectives and specific services offered through the ORA Quality Improvement and Outreach Program. Details regarding program components are outlined in SOP-801 (Post-Approval Support) and SOP-802 (Post-Approval Verification).

Authority

Federal regulations [NASA 14 CFR 1230.109] afford the IRB authority to approve, require modifications, disapprove, suspend, or terminate human subjects research activities at NASA. Additionally, 14 CFR 1230.109(g) specifies that the IRB, or a third party, has the authority to observe the consent process and the research. The program operates under the Office of the Chief Health and Medical Officer (OCHMO)'s Office of Research Assurance in support of the NASA IRB.

Objectives

The objectives of NASA's Quality Improvement and Outreach Program (QIOP) include providing:

Study level

- An internal verification process to confirm human subjects research projects are conducted in accordance with Institutional Review Board (IRB)-approved protocols.

Research Team level

- Practical support, guidance, and education related to conduct of HSR that facilitates research and optimizes compliance.
- A collegial contact within the Office of Assurance
- Identification of strengths and best practice for conduct of HSR
- Targeted research team education to address identified gaps.

Center level

- A means to identify systemic education needs related to conduct of HSR
- Harmonization and consistency across Centers

IRB level

- Promote consistency across reviews.
- Internal checks for documentation accuracy and consistency
- Ensure IRB practices and reviews align with regulatory requirements.

External IRBs with PI engaged in collaborative research with NASA.

- A means for assessing compliance and providing oversight of studies.

Roles

IRB

The NASA IRB is an Agency-wide IRB operating within the Common Rule [NASA 14CFR1230] charged with oversight of research involving human subjects to ensure the ethical, safe, and equitable treatment of human subjects. The NASA IRB has the authority to approve, require modifications, disapprove, suspend, or terminate human subjects research activities. As part of the IRB's role, the Common Rule explicitly grants IRBs the authority to observe, or have a third party observe, the research and consent process.

The IRB is administratively supported by the Office of Research Assurance (ORA), within the Office of the Chief Health and Medical Officer (OCHMO) Health Operations and Oversight Directorate.

Director, Office of Research Assurance (ORA)

The Director of Office of Research Assurance (ORA) provides support and management of the NASA IRB and HRMRB, office (IRB Lead and IRB and HRMRB coordinators) and the ORA Quality Improvement Outreach Lead.

ORA Quality Improvement and Outreach (QIO) Lead

The ORA QIO Lead's role is to provide support to the NASA research community through outreach, education, and the proactive identification of potential compliance concerns.

As part of the verification process, the QIO Lead seeks to confirm through observation and document comparison that research is performed in accordance with IRB approved protocols. The QIOP Lead reviews study records, observes research activities, prepares reports, provides recommendations for maintaining research compliance, and assists with the development of plans for corrective and preventative actions as needed.

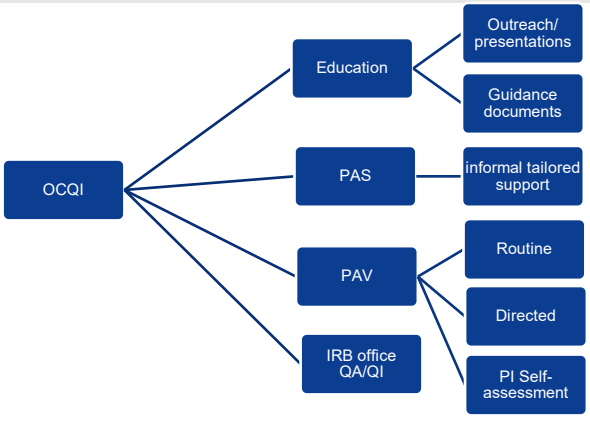
The Quality Improvement and Outreach Lead also develops and delivers guidance and educational programming to the NASA research community.

Principal Investigators and research teams

Principal Investigators (PI) and research teams hold the most critical role in ensuring that human subjects are protected, and that research adheres to regulatory requirements. PI and research team members are responsible for ensuring that HSR activities receive prior IRB review and approval, including any proposed modifications or revisions to previously approved research.

Services

The ORA Compliance and Quality Improvement QIO) program offers a variety of services to support investigators and research staff in the conduct of ethical research in compliance with federal regulations and NASA policies.



Education

Educational presentations and outreach to the NASA research community is a critical component of the QIO program.. Presentations on a variety of topics are scheduled throughout the year. Upon request, staff conduct any size group training on topics related to Human Subjects Research (e.g., federal regulations, NASA policy, documentation, informed consent). Additionally, Post-Approval Support and Post-Approval Verification visits allow for one-on-one individualized information for PI and research teams. QIO staff also work to develop IRB guidance and study management tools to facilitate research in support of NASA's mission.

Post-Approval Support (PAS)

Post-Approval Support (PAS) is an initiative to provide direct and tailored support to research teams in maintaining compliance over the lifetime of a research project. Following a brief initial tag to assess needs, timelines, availability, a PAS plan is developed by QIO staff. The plan includes proposed services catered for the study (i.e., study start-up, documentation best practices, check-in points, verification/compliance assessments etc.). Services are intended to be educational and serve as touch points for proactive correction (if needed).

Post-Approval Verification (PAV)

All studies under oversight of the NASA IRB, including studies with oversight ceded to an external IRB may be selected for verification through the PAV program. Verification may fall under one of the following categories:

Routine Verification

Routine Verification (Not for Cause) – Studies chosen for **Routine Verification** are primarily randomly selected; however, emphasis may be placed on studies involving:

- Greater than Minimal Risk procedures,
- Food and Drug Administration (FDA) oversight,
- Spaceflight,
- PI with past compliance issues,
- Subject complaints

The aim of Routine Verification is to confirm compliance, proactively identify, and if necessary, rectify potential problems. Routine Verification typically involves an in-depth review of the overall study; however, more targeted checks (e.g., consent documentation review, observations of consent process, review of data security provisions) may occur.

Directed Verification

Directed Verification (For Cause)- conducted at the request of the IRB or IRB Chair.

Performed when:

- There are concerns regarding compliance, protocol adherence, participant safety or well-being,
- Confirmation of corrective action implementation is needed,
- Requested by external institutions in the context of a Reliance Acknowledgment (RA)

PI Self-Assessment

At any time, a PI may choose to complete a self-assessment using the PI Self-Assessment checklist. Completion of a self-assessment facilitates timely identification and correction of potential compliance concerns.

IRB Office QA/QI

QIO staff also provide quality check information for the ORA IRB office. File audits to ensure accurate internal documentation (e.g., eIRB study file, minutes, etc.) may be conducted randomly or in tangentially with a study PAV. Additionally, OCQI staff gather and analyze IRB office metrics related to volume and processing times to promote a culture of continued process improvement.

Acronyms

FDA – Food and Drug Administration

HSR – Human Subjects Research

IRB – Institutional Review Board

OCHMO – Office of the Chief Health and Medical Officer

OQIO– ORA Quality Improvement and Outreach

ORA – Office of Research Assurance

PAS – Post-Approval Support

PAV – Post-Approval Verification

PI – Principal Investigator

QIOP – Quality Improvement and Outreach Program

RA – Reliance Acknowledgement

RNI – Report of New Information

QA/QI – Quality Assessment/Quality Improvement

Version #	Version Notes:	Originated By:	Approved By:	Date Approved:
v.1.1	Update program name, typographical	J. Ensley Gorshe	M. Covington	19Feb2025
v.1	Original	J. Ensley Gorshe	M. Covington	24Jul2024